

Logistic regression interpretation

Ciaran Evans

Recap: logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Recap: interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Intercept:

- + when age is 0, the estimated log odds of dengue is -2.45
- + when age is 0, the estimated odds of dengue is $e^{-2.45} = 0.086$

Recap: adding more variables

Now let's add WBC as a variable to the model:

```
m2 <- glm(Dengue ~ Age + WBC, data = dengue,  
          family = binomial)  
summary(m2)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 0.34 + 0.15 \text{ Age}_i - 0.31 \text{ WBC}_i$$

A one-year increase in age is associated with an increase in the odds of dengue by a factor of $e^{0.15} = 1.16$, holding white blood cell count fixed.

Adding more variables

What about a *categorical* predictor?

= 1 if pain
= 0 if no pain

```
m3 <- glm(Dengue ~ Age + Abdominal, data = dengue,  
           family = binomial)  
summary(m3)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -0.25 + 0.22 \text{ Age}_i + 0.21 \text{ Abdominal}_i$$

How should I interpret the coefficient on abdominal pain?

The presence of abdominal pain is associated with an increase in odds of dengue by a factor of $e^{0.21}$ (compared to patients with no pain), holding Age fixed

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Can I interpret the coefficient in terms of the *probability* (rather than odds or log odds)?

Short answer: No

$$1.25 = e^{0.22}$$

	<u>Age = 10</u>		<u>Age = 11</u>		<u>Age = 12</u>
log odds	-0.25	$\xrightarrow{+0.22}$	-0.03	$\xrightarrow{+0.22}$	0.19
odds	0.78	$\xrightarrow{\times 1.25}$	0.97	$\xrightarrow{\times 1.25}$	1.21
$\hat{\pi}_i$	0.44	$\xrightarrow[\times 1.1]{+0.05}$	0.49	$\xrightarrow[\times 1.12]{+0.06}$	0.55

different!

- constant additive change in log odds
- constant multiplicative change in odds
- no constant change in probability